

CODECLASH Problem Statements

Agriculture, FoodTech & Rural Development	Analyzing data derived from the Govt. Land Information System (GLIS) : Harnessing GLIS Data for Societal Challenges: Develop analytics solutions utilizing GLIS data to address societal challenges and inform evidence-based decision-making. Solutions should focus on domains like Urban Planning, Infrastructure Development, Environmental Conservation, Land Management, and Socio-economic Analysis.
Blockchain & Cybersecurity	Developing a system for generating and validating certificates using blockchain technology for government entities : Blockchain-Based Certificate Generation and Validation: Create a system for custom digital certificate generation and validation using open-source software and blockchain technology. Expected output includes certificate integration with a Digital Locker System, catering to Government Offices, Students, Industries, and Institutes.
Blockchain & Cybersecurity	Monitoring and ensuring compliance with security measures in cloud environments : Cloud Environment Monitoring and Auditing Tools: Develop software tools for monitoring and auditing cloud environments to detect security incidents, compliance violations, and anomalous activities. Features should include log analysis, intrusion detection, and security event correlation.

Blockchain & Cybersecurity	Creating an intelligent system utilizing AI/ML to identify phishing domains mimicking genuine websites : AI/ML-Based Phishing Domain Detection: Implement an automated tool utilizing AI/ML to identify phishing domains from genuine ones based on open-source databases like WHOIS. Techniques include code/content similarity analysis and web page image analysis for probability scoring of phishing domains.
Clean & Green Technology	Engaging students in innovative projects and initiatives : Waste Management and Sanitization Solutions: Propose solutions for waste segregation, disposal, and sanitation system improvements.
Clean & Green Technology	Developing a mobile application aimed at promoting environmentally friendly practices : Pro Planet Mobile Application: Design a mobile application to track user activities, promote eco-friendly behaviors, and suggest sustainable alternatives in real-time across various applications like food delivery and online ordering.
Disaster Management	Creating a mobile app that gathers and displays water-related issues within communities through crowdsourcing and mapping : Social Media Data Utilization in Disaster Management: Develop software/algorithms to analyze ground-level images shared on social media for disaster management, focusing on water-related problems like flooding and drainage issues.
Disaster Management	Establishing an application where rescue agencies register and coordinate their efforts, particularly during disasters : Rescue Agency Coordination Application: Create a centralized database and application for registering

	and locating rescue agencies to facilitate coordination and aid distribution during natural or man-made disasters.
FinTech	Exploring platforms for decentralized finance, potentially disrupting traditional financial systems : Decentralized Finance (DeFi) Platforms: Build decentralized finance applications leveraging cryptocurrency and blockchain for innovative financial services.
FinTech	Predicting trends in various markets using data analysis and forecasting techniques : Market Trend Prediction Tools: Analyze historical and real-time market data to predict trends and optimize investment decisions.
MedTech / BioTech / HealthTech	Designing software to aid in suggesting drugs and formulations based on Ayurvedic classical texts and repositories : Speech Therapy/Education Software for Neurodivergent Individuals: Develop software for speech therapy or education tailored for neurodivergent individuals.
MedTech / BioTech / HealthTech	Developing software to recommend medications and formulations for specific diseases or pharmacological properties based on Ayurvedic classical literature and repositories. : Ayurvedic Drug and Formulation Identification Software: Create custom software to identify Ayurvedic drugs and formulations tailored to specific symptoms and conditions, including pharmacological properties and contraindications.

Tourism	Digital Tourism Solutions for Ayodhya : Digital Tourism Enhancement Solutions for Ayodhya: Leverage digital technologies like mobile apps, VR, AR, and geospatial mapping to enhance tourism experiences, promote Ayodhya as a tourist destination, and facilitate online bookings and navigation.
Tourism	Develop a software application that leverages advanced technology to address key challenges and enhance the tourism experience in India. The application should provide innovative solutions to improve accessibility, promote cultural immersion, and streamline travel logistics for both domestic and international tourists : Developing an innovative tourism app for India, aiming to overcome infrastructure, language, and information accessibility challenges. The solution integrates AR tours, real-time translation, and personalized trip planning to enhance cultural immersion and streamline travel logistics. By addressing these barriers, the app aims to revolutionize the tourism experience and position India as a leading global destination.
Miscellaneous	Creating immersive augmented reality (AR) experiences for various applications : Augmented Reality (AR) Application Development: Explore AR technology by developing Android apps overlaying digital content onto the physical world, fostering innovation and enhancing user experiences.
Miscellaneous	Ensuring the robustness and reliability of algorithms through rigorous testing and validation processes : Robust Algorithm Design: Design algorithms resilient to outliers, noise, missing

	data, and adversarial attacks, emphasizing robust optimization and statistics techniques.
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